Neerav Chaudhary

(669)249-0652 | neerav.ch@icloud.com | LinkedIn | GitHub | Portfolio

SUMMARY

Engineering Management graduate and AI engineer building conversational systems that turn complex operational and research workflows into clear, actionable experiences. Combining voice interfaces, agentic automation, and product design with human-in-the-loop controls to help users resolve blockers, complete approved actions, and interact safely with connected tools, data, and workflows.

PROJECTS

AI-Native Employee Onboarding & Assistance Portal | Deepgram
Voice Agent, TypeScript, Slack

- Built an AI-first onboarding and employee-assistance portal spanning 7+ workflows across tasks, learning, documents, equipment, benefits, payroll, and support cases. Employees can view status, deadlines, blockers, and next actions, then converse with the Deepgram agent by voice or text for onboarding support
- Developed the real-time voice layer with Deepgram Nova-3 for speech recognition, the Browser Agent SDK for session orchestration and interruption handling, structured function calling for authenticated data retrieval and workflow execution, and Flux TTS for spoken responses.
- Connected 5 controlled functions and 3 executable actions to let employees use voice to start tasks, schedule reminders, create support cases, and escalate blockers across IT, Administration, Finance, and HR. The system identifies the issue, routes it to the correct owner, and guides the employee through the next step, reducing manual People-team coordination for access, equipment, payroll, and onboarding requests.
- Implemented role-based access controls and employee confirmation for task initiation, support-case creation, and reminder scheduling, with short-lived credentials, duplicate-action prevention, and append-only audit logs. Validated the system across a 13-step end-to-end workflow covering interruptions, failed actions, recovery, and policy enforcement
- Built a Slack integration for onboarding follow-up, using identity-mapped and signed workflows to deliver reminders, route escalations, and collect manager approvals. Added duplicate-request protection so employees and managers could complete follow-up in Slack while preserving the same authorization and audit controls as the portal.

Refract-AI: Research & Predictive-Maintenance Platform **Jan 2026 - May 2026**

- Built a no-code statistical analysis and predictive-maintenance interface that takes raw lithium-ion battery data through data-quality checks, leakage screening, feature analysis, hypothesis testing, Multiple Linear Regression, Random Forest comparison, and Matplotlib visualizations in one repeatable workflow; removed the need to assemble separate analysis steps and validated Random Forest RUL at 25.6-cycle RMSE.
- Integrated RAG literature synthesis with statistical and predictive outputs in the same interface, reducing end-to-end research time 45% and enabling structured comparison of 10+ technical papers per session; connected model results to evidence on battery degradation, second-life applications, renewable-energy storage, grid reliability, and energy-hardware research.
- Added an OpenAI Voice-powered research coach that lets users discuss a paper conversationally, ask follow-up questions, and recite their understanding aloud; the system compares the explanation with the source paper and returns accuracy-focused feedback, turning literature review into an interactive comprehension and knowledge-check workflow.
- Generated reusable validation and benchmarking reports that surfaced 8+ research gaps and identified 15+ comparable datasets and studies; packaged assumptions, model results, and supporting evidence into repeatable outputs for engineering review across BESS, grid, renewable-energy, and data-center reliability use cases.

PROFESSIONAL EXPERIENCE

Deep Strat | Artificial Intelligence Research Intern **Aug 2023 - Nov 2023**

- Partnered with department leadership to define customer-feedback and complaint-handling requirements; built a 3-stage LLM workflow that classified sentiment and intent, retrieved routing context to assign each message to the responsible department, and generated SOP-grounded reports so teams received structured, reviewable cases instead of raw customer messages.
- Authored a 50-page technical reference synthesizing AI-safety research, model failure modes, debates among leading researchers, and original analysis; organized findings by recurring failure patterns, competing research positions, and areas requiring further investigation, then presented the reference to company founders, the COO, and research assistants. The work supported jointly authored AI-safety research presented at Nullcon.
- Evaluated Random Forest and CNN models on 2,000 customer records against a cost-sensitive objective tied to operational profit; reduced churn-model false positives 25% while maintaining stable AUC, decreasing unnecessary intervention flags without weakening overall model discrimination.

Manipal University Jaipur | AI Researcher, Medical Imaging **Aug 2022 - Dec 2022**

- Improved dementia-severity classification accuracy from 78% to 85% through CNN ensemble comparison and error analysis on ADNI MRI data, strengthening model selection with measured classification performance.
- Built a repeatable Python/OpenCV pipeline for MRI preprocessing and brain-atrophy quantification, converting scans into consistent structural measurements for model evaluation and supporting the published IEEE study on computer-vision-based dementia analysis.

EDUCATION

San Jose State University **Aug 2024 - May 2026**
M.S., Industrial / Engineering Management (GPA: 3.81)

Manipal University Jaipur **Aug 2018 - Jun 2022**
Bachelor's, Computer Engineering (GPA: 8.8)

PUBLICATIONS

N. Chaudhary, S. S. Verma - Analyzing Brain Atrophy in Dementia Using Computer Vision. IEEE ICRITO, 2022.
Published computer-vision research applying image preprocessing and structural quantification to MRI scans to measure brain atrophy associated with dementia.

SKILLS

AI & Agentic Systems: Deepgram Nova-3, Browser Agent SDK, Flux TTS, OpenAI Voice, voice agents, conversational AI, structured function calling, RAG, LLM workflows, multi-agent systems, human-in-the-loop controls, model evaluation, error analysis, validation.
Development & Integrations: TypeScript, Python, SQL, R, React, Slack workflows/APIs, REST APIs, API orchestration, FastAPI, Express, SQLite, PostgreSQL, AWS, Docker, Git, authentication/authorization, idempotency, audit logging.
ML, Research & Operations: Hypothesis testing, regression, Random Forest, CNNs, data-quality checks, leakage screening, Matplotlib, evidence synthesis, literature review, technical writing, requirements gathering, stakeholder collaboration, Microsoft Excel, Lean Six Sigma.
Product & Design Engineering: Internal tools, workflow and interaction design, voice UX, conversational workflows, user flows, rapid prototyping, requirements translation, human-centered automation, People Operations automation, cross-functional product development.